# Initial Project Planning Template

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| **Date** | 03 July 2026 |
| **Team ID** | N/A |
| **Project Name** | OptiCrop – Smart Agricultural Production Optimization Engine |
| **Maximum Marks** | 5 Marks |

## Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product backlog and sprint schedule for building OptiCrop.

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members | Sprint Start Date | Sprint End Date (Planned) |
|---|---|---|---|---|---|---|---|---|
| Sprint-1 | Data Collection & Preprocessing | USN-1 | As a researcher, I can load and explore the crop dataset (N, P, K, temperature, humidity, pH, rainfall) so that I understand its structure before modeling. | 2 | High | — | 23 Jun 2026 | 26 Jun 2026 |
| Sprint-1 | | USN-2 | As a researcher, I can visualize feature distributions (univariate/bivariate/multivariate) so that I can understand data patterns before modeling. | 2 | High | — | 23 Jun 2026 | 26 Jun 2026 |
| Sprint-1 | | USN-3 | As a researcher, I can clean null values and handle outliers (e.g., in Potassium) so that the dataset is reliable for training. | 2 | Medium | — | 23 Jun 2026 | 26 Jun 2026 |

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|---|---|---|---|---|---|---|---|---|
| Sprint-1 | | USN-4 | As a developer, I can split the data into training and testing sets so that model performance can be evaluated fairly. | 1 | High | — | 23 Jun 2026 | 26 Jun 2026 |
| Sprint-2 | Model Building | USN-5 | As a developer, I can train multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest) so that I can compare their performance. | 3 | High | — | 27 Jun 2026 | 30 Jun 2026 |
| Sprint-2 | | USN-6 | As a developer, I can compare model performance using accuracy, precision, recall, and F1-score so that I can select the best model. | 2 | High | — | 27 Jun 2026 | 30 Jun 2026 |
| Sprint-2 | | USN-7 | As a developer, I can save the best-performing model and scaler using Pickle so that predictions don't require retraining. | 1 | High | — | 27 Jun 2026 | 30 Jun 2026 |
| Sprint-2 | | USN-8 | As a researcher, I can explore agricultural condition clusters using K-Means so that I can identify natural groupings in the data. | 2 | Low | — | 27 Jun 2026 | 30 Jun 2026 |
| Sprint-3 | Application Building | USN-9 | As a farmer, I can view the OptiCrop home and about pages so that I understand what the tool does. | 1 | Medium | — | 01 Jul 2026 | 03 Jul 2026 |
| Sprint-3 | | USN-10 | As a farmer, I can enter soil and climate values in a form so that I can request a crop recommendation. | 2 | High | — | 01 Jul 2026 | 03 Jul 2026 |
| Sprint-3 | | USN-11 | As a farmer, I can get an instant crop recommendation after submitting the form so that I can decide what to plant. | 3 | High | — | 01 Jul 2026 | 03 Jul 2026 |

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|---|---|---|---|---|---|---|---|---|
| Sprint-3 | | USN-12 | As a farmer, I want the interface to be clean and easy to use on any device so that I can use it without technical difficulty. | 2 | Medium | — | 01 Jul 2026 | 03 Jul 2026 |